1 Executive Summary

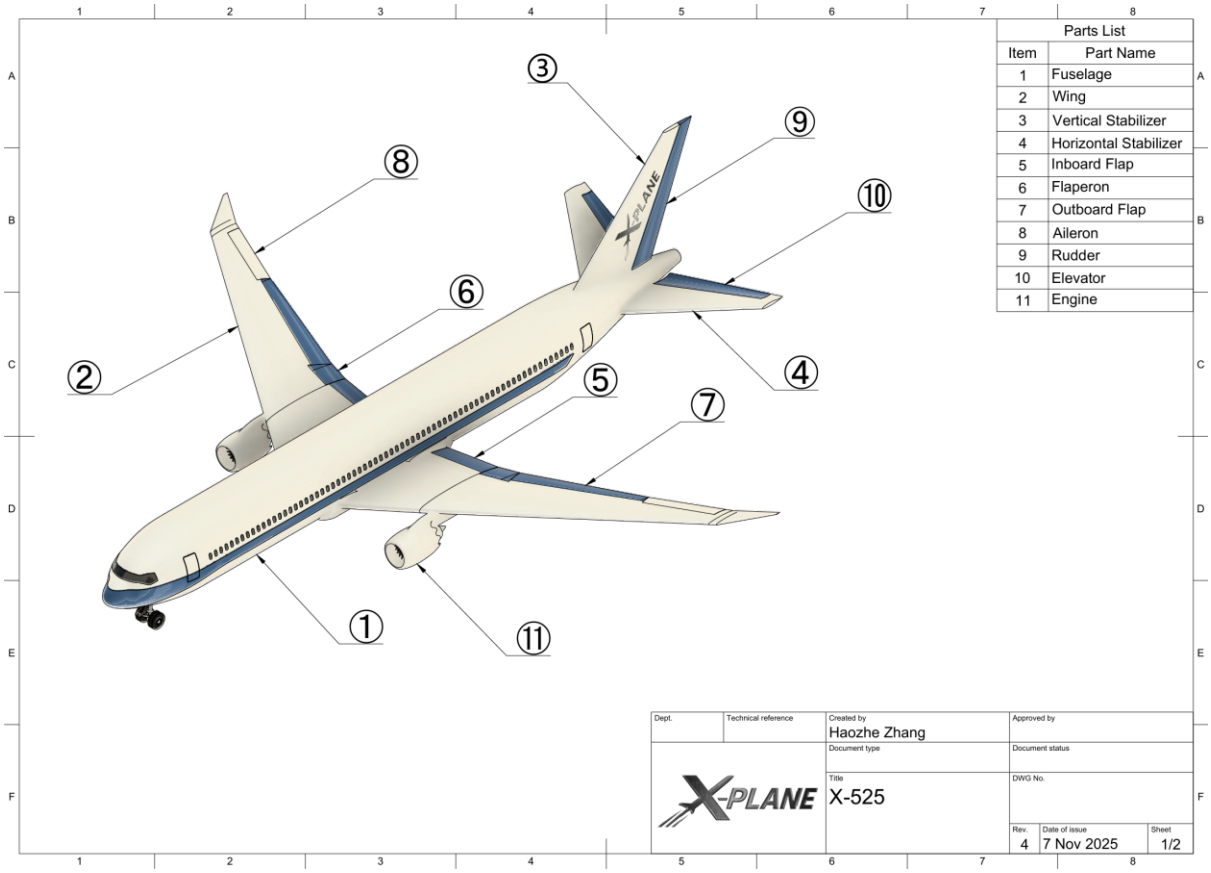


Figure 1: Isometric view of X-525

2 Aircraft Sizing

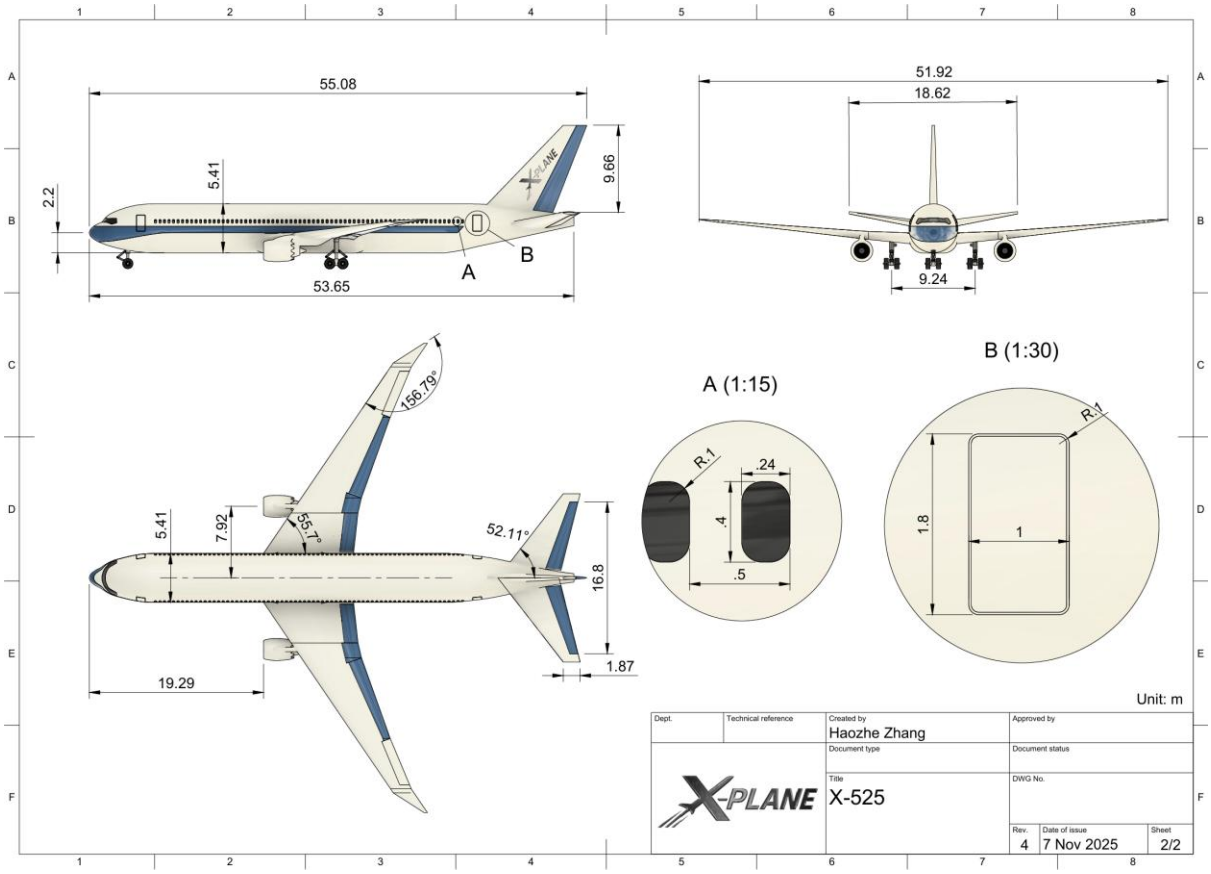


Figure 2: Three-view drawing of X-525

3 Wing Design

3.1 Wing Control Surfaces



Figure 3: Fully deployed flaps and ailerons

3.2 Structural Configuration

3.2.1 Material Selection

Aluminum alloy AL2024-T42 is selected as the airframe material, with the material properties shown in Table 1. This alloy is widely employed in aerospace applications due to its high strength-to-weight ratio and good fatigue resistance [1] which make it particularly well-suited for aircraft structures subject to cyclic loading and potential impact damage.

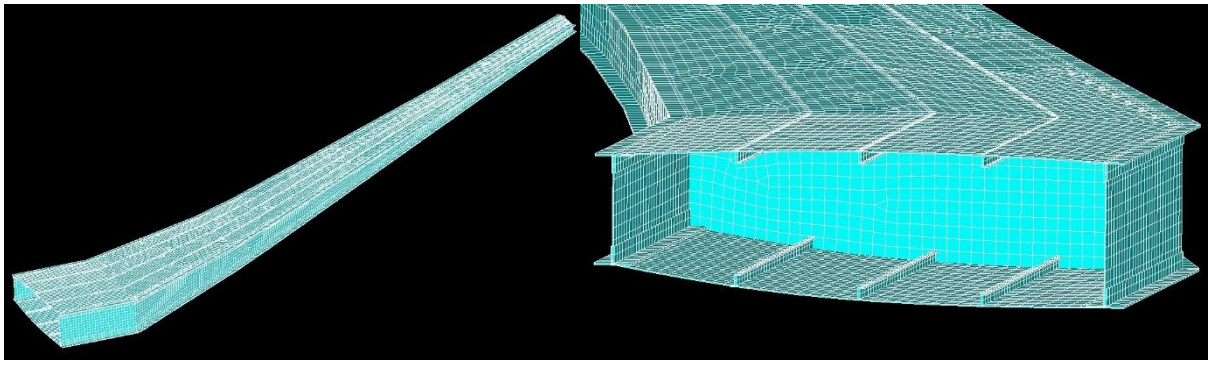
Table 1: Fuselage structural component dimensions

2024-T42 Aluminum	
Density	2.77 g/cm^3
Young's Modulus	72.4 GPa
Shear Modulus	28.0 GPa
Poisson's Ratio	$\leq$ 0.33
Yield Strength (Tensile)	$\geq$ 250 MPa @Thickness 1.60 - 12.7 mm
Ultimate Strength (Tensile)	$\geq$ 415 MPa @Thickness 1.60 - 12.7 mm

3.2.2 Structural Dimensioning

The wing primary structure consists of a two-spar configuration with the front spar positioned at 25% chord from the leading edge and the rear spar located at 35% chord from the trailing edge. Between the spars, three stringers are evenly distributed on both the upper and lower wing skins to provide longitudinal stiffening. The spars and stringers utilize T-section and Z-section profiles respectively. The skin thickness is 10 mm.

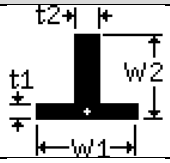
The finite element mesh of the left wing structure and the corresponding spars and stringers are illustrated in Figure 4, while the geometric dimensions of the primary structural components are summarized in Table 2.



(a) (b)

Figure 4: Mesh elements of (a) left wing structure and (b) spars and stringers

Table 2: Wing structural component dimensions

Component	Cross Section	Dimensions (mm)					
Spar		W1	W2		t1		t2
		300	150		15		30
Stringer		W1	W2	W3	t1	t2	t3
		50	50	100	10	10	20

3.3 Structural Analysis

The continuous spanwise load distribution acting on the wing structure under normal flight conditions is illustrated in Figure 5.

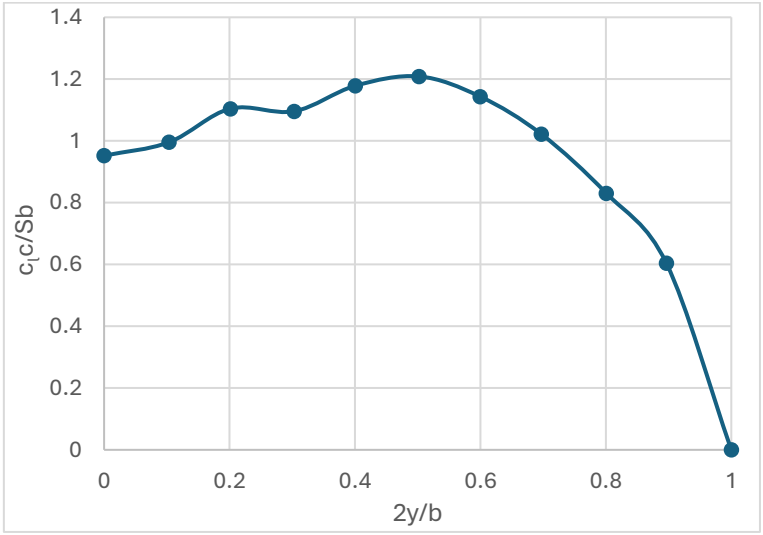


Figure 5: Continuous spanwise load distribution curve

To analyze the deformation and stress response of the wing at an acceleration of 2.5g, the continuous aerodynamic load was discretized into concentrated loads located at the rib positions, as shown in Figure 6. The corresponding magnitudes of the simplified loads are listed in Table 3. This discretization method significantly enhances computational efficiency while maintaining adequate accuracy in the finite element simulation results.

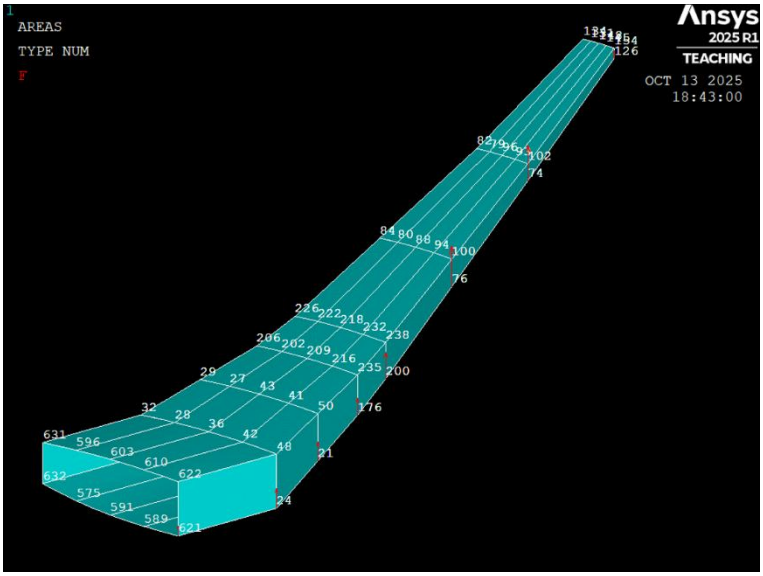


Figure 6: Locations of simplified concentrated loads at each rib

Table 3: Magnitudes of simplified concentrated loads at 2.5g

Point #	**621**	**24**	**21**	**176**	**200**	**76**	**74**	**126**
Lateral Distrance from Center (b)	0	0.12	0.21	0.31	0.38	0.56	0.76	1
Load (kN)	118	235	224	200	315	482	420	91

The finite element analysis (FEA) was conducted using ANSYS Mechanical APDL, incorporating all major structural components - spars, stringers, ribs, and skin panels - to evaluate the wing's performance under the specified loading condition. Boundary conditions were defined to replicate the fuselage attachment constraint, with the wing root section fully fixed in all degrees of freedom.

The deformation results of the left wing at 2.5g are presented in Figure 7. The analysis indicates a maximum tip displacement of 2.733 m, which lies within acceptable limits. The deformation pattern exhibits the typical cantilever bending behavior, with deflection progressively increasing from the wing root toward the tip.

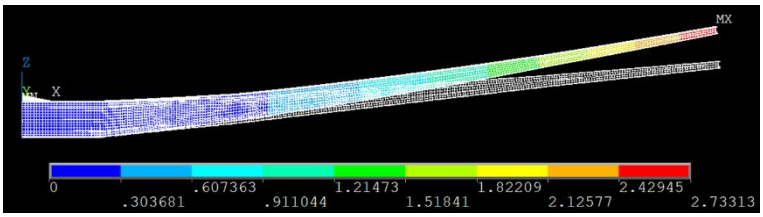


Figure 7: Deformation of the left wing at 2.5g (front view)

The von Mises stress distribution of the left wing is illustrated in Figure 8. The stress contours indicate highest stress levels in the inboard region, corresponding to areas of maximum bending moment. The color gradient enables clear identification of critical stress regions that warrant further detailed design evaluation.

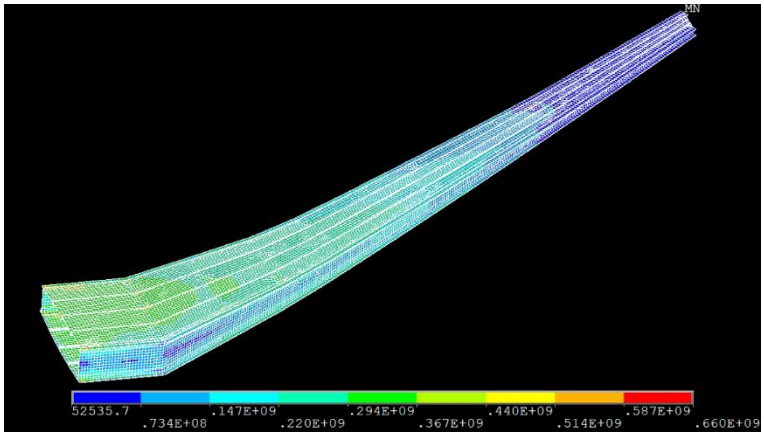


Figure 8: Stress distribution of the left wing at 2.5g loading

The maximum stress occurs at the top corner of the front spar within the wing box, near the plane of symmetry at the fuselage interface, as shown in Figure 9. This region experiences combined effects of maximum bending moment and geometric stress concentration. The peak stress value at this critical location should be compared against the material allowable stress, applying suitable safety factors to ensure structural integrity.

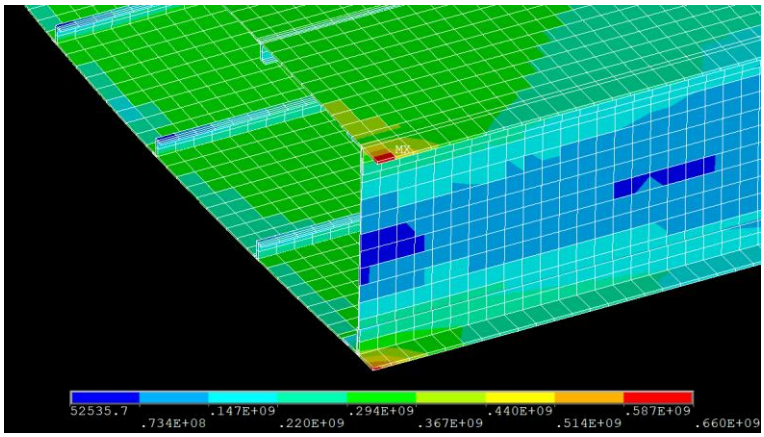


Figure 9: Maximum stress of left wing at 2.5g loading

4 Fuselage Design

4.1 Structural Configuration

The fuselage structure of the X-525 aircraft comprises three primary load-bearing components: frames, longerons, and skin panels. These elements act together to ensure the required stiffness, strength, and pressurization integrity of the fuselage shell.

4.1.1 Material Selection

Material selection for the X-525 airframe is governed by strength-to-weight efficiency, fatigue life, and corrosion resistance, ensuring structural integrity throughout the aircraft's service life. The chosen materials must satisfy FAR Part 25 safety requirements while minimizing operating and manufacturing costs.

Accordingly, the design utilizes lightweight, high-strength alloys and advanced composites to achieve optimal performance within weight and cost constraints. Emphasis is placed on materials that provide a favourable stiffness-to-density ratio, excellent fatigue and damage tolerance, and corrosion resistance, ensuring long-term durability and maintainability across all major load-carrying components.

The chosen materials for each fuselage component are summarized in Table 4.

Table 4: Material choices for fuselage components

Component	Material	Young's Modulus	Poisson's Ratio
Skin	Carbon-Fibre Fabrics [3]	70 GPa	0.1
Frame	2024-T42 Aluminum	72.4 GPa	0.33
Longeron			

4.1.2 Structural Dimensioning

Based on the selected materials, the dimensional configuration of the fuselage structural components was determined following basic design guidelines and simplified structural optimization methods [4][5]. The resulting dimensions for the frames and longerons are summarized in Table 5.

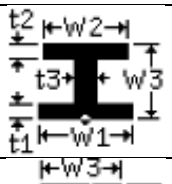
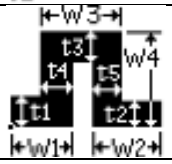
Frames provide the circumferential stiffness necessary to maintain the fuselage geometry and effectively transmit internal pressure and bending loads into the longitudinal members. They are evenly spaced at 1 m intervals along the fuselage length.

Longerons, positioned at 15° intervals around the fuselage circumference (a total of 24 longerons), serve as the primary longitudinal load paths, carrying bending and axial loads generated during pressurization and flight maneuvers.

The fuselage skin acts as a primary load-bearing element, resisting pressurization-induced stress. In this model, the skin thickness is specified as 10 mm.

Table 5: Fuselage structural component dimensions

Component	Cross Section	Dimensions (mm)					
Frame		W1	W2	W3	t1	t2	t3

		50		50	100		10	10		15
Longeron		W1	W2	W3	W4	t1	t2	t3	t4	t5
		25	25	50	50	10	10	10	10	10

Based on the above details, the frames and longerons were modelled and assembled to form the fuselage's structural support framework, as illustrated in Figure 10.

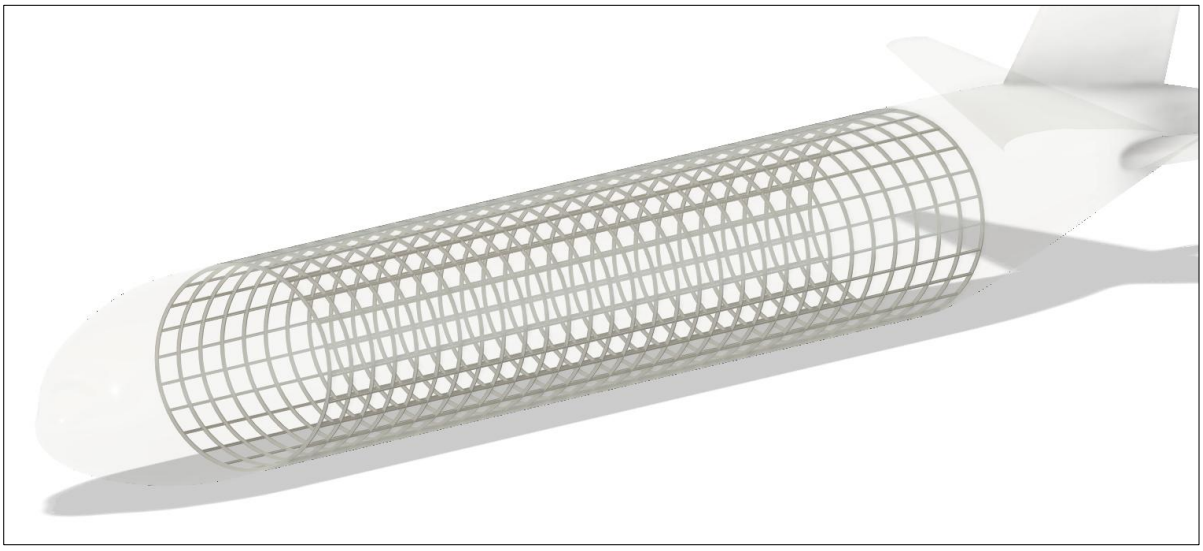
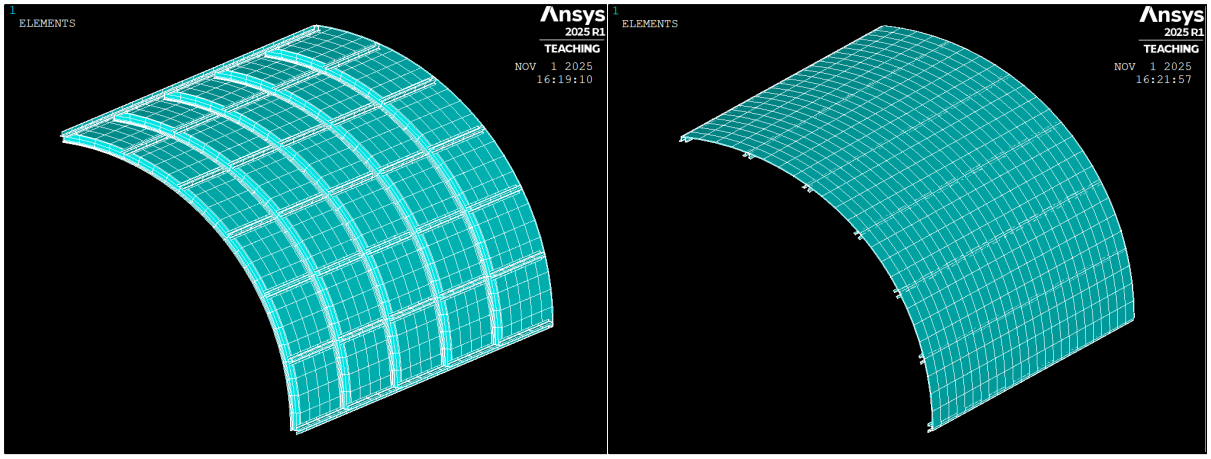


Figure 10: Frames and longerons

4.2 Structural Analysis

A structural analysis of the fuselage under pressurization was conducted using ANSYS Mechanical APDL. To simplify the computation and leverage geometric symmetry, only a quarter cross-section of the fuselage, with a length of 5 m, was analyzed. Two planes of symmetry were applied to represent the full structure. The meshed model, shown in Figure 11, was generated using a global element size of 0.2 m, with both inner and outer mesh views displayed.



(a) (b)

Figure 11: (a) Inner view and (b) outer view of mesh elements of quarter fuselage model

Following meshing, boundary conditions were applied to reflect the symmetric loading conditions. The tangential displacement along the two symmetry planes of the cylindrical fuselage was constrained, permitting motion only in the radial and axial directions.

The analysis considered pressure-induced stresses as the dominant load case. An internal pressure was applied normal to the fuselage skin shell elements to simulate the hoop stress, while the longitudinal stress was carried jointly by the skin and longitudinal beam elements.

The resulting displacement vector sum is presented in Figure 12. The analysis indicates a maximum displacement of less than 0.6 mm, which is within acceptable limits for fuselage structural integrity.

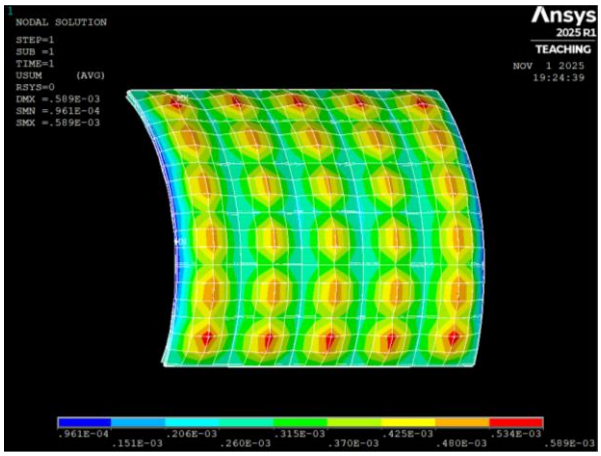


Figure 12: Displacement vector sum of the fuselage section

5 Empennage Design



Figure 13: Empennage



Figure 14: Elevators and maximum deflection



Figure 15: Rudder at maximum deflection

6 References

[1] M. Kerster, “2024-T42 and natural age,” AAA Air Support, https://www.aaaairsupport.com/2024-t42-and-natural-age/#:~:text=2024%2DT42%20is%20a%20specific,Wilm%20at%20D%C3%BCrener%20Metallwerke%20AG. (accessed Oct. 15, 2025).

[2] Alclad aluminum 2024-T42, https://www.matweb.com/search/DataSheet.aspx?MatGUID=d0ab706c407443078a647eafcbce9b7e (accessed Oct. 15, 2025).

[3] “Mechanical properties of carbon fibre composite materials, fibre / epoxy resin (120°C cure),” Mechanical Properties of Carbon Fibre Composite Materials, https://www.performance-composites.com/carbonfibre/mechanicalproperties_2.asp (accessed Nov. 4, 2025).

[4] ‘Fig. 1. Cross-section of a typical compression skin-stringer panel.’, ResearchGate. Accessed: Nov. 02, 2025. [Online]. Available: https://www.researchgate.net/figure/Cross-section-of-a-typical-compression-skin-stringer-panel_fig3_222649676

[5] C.-S. Chen, P. A. Wawrzynek, and A. R. Ingraffea, ‘Prediction of Residual Strength and Curvilinear Crack Growth in Aircraft Fuselages’, *AIAA Journal*, vol. 40, no. 8, pp. 1644–1652, Aug. 2002, doi: 10.2514/2.1836.